SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Department of Computer Science and Engineering

ITA0303-Mobile Computing

Real-Time Food Delivery Mobile App

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What We're Building

A full-featured food delivery platform connecting customers, restaurants, and delivery partners — with live GPS tracking, secure payments, and intelligent notifications at every step.

🍽️ Food Selection

Browse menus, customize orders, and manage your cart effortlessly.

💳 Online Payment

Secure, integrated payment gateways for a frictionless checkout.

📍 Live GPS Tracking

Real-time order status and delivery partner location on Google Maps.

🔔 Push Notifications

Instant alerts for order updates, driver arrival, and promotions.

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ARCHITECTURE OVERVIEW

Three Core Modules

The app is structured into three tightly integrated modules, each handling a distinct phase of the delivery lifecycle — from onboarding to final delivery.

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Module 1

User & Restaurant Management

2

Module 2

Order Processing & Real-Time Tracking

3

Module 3

Delivery & Admin Management

MODULE 1

User & Restaurant Management

Customer Side

- User registration, login, and profile management
- Search and filter restaurants by cuisine, rating, and distance
- View detailed menus with images, prices, and availability
- Save favorite restaurants and reorder history

Restaurant Side

- Restaurant onboarding and profile setup
- Menu management — add, edit, or remove items in real time
- Order acceptance/rejection dashboard
- Business hours and availability toggle

MODULE 2

Order Processing & Real-Time Tracking

From cart to confirmation — every step is tracked, updated, and communicated in real time.

Browse & Add

Select items and add to cart

Checkout & Pay

Complete order with online payment

Order Confirmed

Restaurant notified automatically

Live GPS Tracking

Real-time location updates

Orders flow seamlessly from selection through payment to live tracking, with each stage triggering automatic notifications to customers and restaurants via the real-time database.

MODULE 2 — DEEP DIVE

Live GPS Tracking & Order Status

Google Maps API Integration

Live map view showing the delivery partner's location, route, and estimated arrival time.

Real-Time Database Sync

Order status updates instantly across all devices — customer, restaurant, and driver.

Push Notifications

Automated alerts for order confirmed, food prepared, driver en route, and delivered.

MODULE 3

Delivery & Admin Management

Delivery Partner Features

- Driver registration, verification, and profile
- Accept or decline delivery requests
- In-app navigation to restaurant and customer
- Delivery completion confirmation with proof

Admin Dashboard

- Monitor all active orders and deliveries
- Manage users, restaurants, and delivery partners
- View analytics: delivery times, ratings, and revenue
- Handle disputes, refunds, and support tickets

Technology Stack

Every tool is chosen to deliver a fast, reliable, and scalable delivery experience.

Android Studio

Primary IDE for building, testing, and deploying the Android app.

Java / Kotlin

Core application logic with modern, concise Kotlin syntax.

SQLite

Local offline database for caching orders and user data.

Google Maps API

Live maps, geocoding, and route optimization for tracking.

BLE Beacons

Bluetooth Low Energy for indoor positioning and navigation.

Push Notifications

Firebase Cloud Messaging for real-time order alerts.

EXPECTED OUTCOMES

What Success Looks Like

Faster Delivery

Optimized routing and real-time coordination reduce average delivery times.

Accurate Tracking

GPS + BLE beacons ensure precise, end-to-end order visibility.

Improved UX

Intuitive UI with XML design delivers a smooth, frustration-free experience.

Higher Satisfaction

Fewer delays, better communication, and transparent tracking boost customer loyalty.

SUMMARY

Key Takeaways

A Complete Delivery Ecosystem

This project delivers a production-ready food delivery app with three tightly integrated modules, covering every stakeholder from customer to admin.

Built for Real-World Performance

- Real-time GPS + BLE indoor positioning
- Offline-capable with SQLite caching
- Secure payments and instant notifications
- Scalable architecture on Android (Java/Kotlin)

Next step: Begin Module 1 sprint — User authentication and restaurant onboarding flow.